Contents

Pre-Registration: Children's inferences about relationships from the way people explain behavior.

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Game URL: https://ashleyjthomas.github.io/RELKIND/ **IRB:** Harvard University Area IRB (approved)

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1. Background

When people explains someone's behavior, they can frame it as a property of the **individual** (e.g., *"Rowan likes to play with bugs"*) or as a property of the **group** the individual belongs to (e.g., *"Wugs like to play with bugs"*). Adults are sensitive to these framing differences and use them to infer the speaker's relationship to the target — people infer social intimacy from individual-level explanations as opposed to group-level explanations or person-level explanations (Thomas et al., in prep)

We ask whether children ages 5 to 8 years reason similarly. Specifically, we test whether children infer social closeness from individual-based explanations, compared to group-based explanations. We contrast this with inferences about hierarchical relationships. We also manipulate whether the speaker already knew the information about the person or seems to be learning it ("Hmm…must" vs. "Yes…"). Adults do not distinguish between these cases suggesting that they think about it as a kind of attention.

Pilot data

We treat the data collected to date (N ~30 participants, ages 4–7) as **pilot data** (any data collected before 6/19/26 2 pm ET). The pilot informed three design decisions:

1. **Excluding 4-year-olds from the registered sample.** Performance and engagement at age 4 in the pilot suggested the task is too verbally demanding for many 4-year-olds (qualitative observations from session videos plus near-chance overall responding at this age).
2. **Adding 8-year-olds.** Older children appeared more sensitive to the manipulation; we extend the upper bound to determine whether the effect grows or plateaus by age 9 years (9 years is now the upper limit).
3. **Planning age-group comparisons.** Patterns in the pilot suggested possible age effects thus we pre-register age analyses and testing our effects in age segmented groups.

2. Hypotheses

H1 — Main effect (best friend block). When answering, *"Who is the [target character]'s best friend?"*, children will be more likely to choose the individual-based explainer than the group-referring speaker (proportion > .5, hypothesis-consistent).

H1 — Main effect (Boss block). When the question is *"Who is the target's boss?"*, children will be less likeliy to choose the individual-based explainer.

H3 — Age effect. Hypothesis-consistent responding will be stronger in older children (ages 7–8) than younger children (ages 5–6).

H4 — Epistemic modulation (exploratory). We will test whether the speaker's expressed certainty (*"Hmm…must"* vs. *"Yes…"*) moderates H1 and H2. In pilot data, the older children, unlike adults only seemed to make the inference about 'Yes' which agrees with prior work suggesting that children infer social closeness from knowing someone else's mental states.

3. Design

A 2 (Question type: Best Friend / Boss) × 2 (Epistemic certainty: Hmm / Yes) within-subjects design over 12 trials. Each child completes one block of each question type (6 trials per block). Within each block, the epistemic frame is held constant and the role assignment (which speaker generalizes vs. individuates) is randomly assigned. Block order, epistemic frame, role assignment, character set (A / B), and left-right position of speakers are counterbalanced across participants. See `README.md` for the full counterbalancing scheme.

4. Sampling plan

Target sample

- **Ages 5 to 8 years**, recruited through Children Helping Science.
- **Target N:** 100 children (25 per year of age), with the goal of at least 80 retained after exclusions (20 per age).
- **Stopping rule:** Recruit until at least 20 retained per year of age (5, 6, 7, 8 ≥ 20 each, after applying exclusion criteria below). If recruitment reaches 120 total

sessions and the target is not met for one age, we will stop and report sample sizes as obtained.
- Stopping rule: If our data are inconclusive we may test more children. Since we are using Bayesian stats we can accumulate evidence for the null.

Inclusion criteria

- Caregiver-reported age 5.00 – 8.99 years on date of session
- Caregiver consented to participate and to data sharing
- Child completed at least one experimental trial
- We will look at 4 year olds as an exploratory investigation

Exclusion criteria

1. Duplicate sessions for the same participant — keep only the first complete or partial session
2. Sessions in which the child was clearly distracted, prompted by the caregiver, or otherwise non-compliant per video review. Whole sessions excluded; individual-trial exclusions are not used.

We will report the number of participants lost at each exclusion step.

5. Variables

Outcome

- Whether children choose the character that provided the person-level explanation or the group-level explanation.

Within-subject manipulations

- question type: Closer (the target is asked about who is best friends) vs. Boss (who is in charge)
- Epistemic: Hmm (tentative) vs. Yes (confident)

Between-subject variables

- `ageGroup`: **Younger (5–6 years)** vs. **Older (7–8 years)**
- age: years (continuous, for sensitivity analyses)

6. Analysis plan

All analyses will be conducted in R using **brms** (Bürkner, 2017) for Bayesian hierarchical models and **BayesFactor** for one-sample comparisons. Code is committed to the project repository.

6.1 Primary confirmatory analysis

We will fit a Bayesian hierarchical logistic regression predicting `hypothesisConsistent` from question type, epistemic frame, age group, and their interactions, with random intercepts (and where feasible, random slopes) by participant:

```
brm(
  answer~ questionType * epistemic * ageGroup + (1|participantId),
  family = bernoulli(),
  prior  = c(
    prior(normal(0, 1.5), class = "Intercept"),
    prior(normal(0, 1),   class = "b"),
    prior(exponential(2), class = "sd")
  ),
  data   = dat,
  iter   = 4000, warmup = 1000, chains = 4, cores = 4,
  control = list(adapt_delta = 0.95)
)
```

Priors. Weakly informative on the log-odds scale: `Normal(0, 1.5)` on the intercept (covers ~5%–95% prior probability on any cell proportion); `Normal(0, 1)` on effect-coded predictors and interactions (mean prior odds-ratio of 1.0, with 95% CrI roughly 0.14–7.4); `Exponential(2)` on random-effect SDs. These priors are conservative — they down-weight extreme effects but are wide enough not to drive inference.

Sum coding will be used for all categorical predictors so that the intercept is interpretable as the grand-mean log-odds and coefficients represent half the difference between levels.

Decision rules. For each effect we will report: - the **95% posterior credible interval** for the coefficient on the log-odds scale and the back-transformed probability scale, - the **posterior probability of the predicted direction** (e.g., $P(\beta > 0 \mid \text{data})$), - the **Bayes factor** for the effect against a null model with that term removed.

We will treat an effect as supported if the 95% CrI excludes zero **and** the directional posterior probability exceeds 0.95. A Bayes factor of **$BF_{10} \geq 3$** (moderate evidence) or **≥ 10** (strong) will be reported as such; **$BF_{10} \leq 1/3$** will be interpreted as evidence for the null.

6.2 Age groupings

Given the pilot pattern we will analyze children in two age groups:

- **Younger:** 5.0 – 6.99 years
- **Older:** 7.0 – 8.99 years

We chose to group 6-year-olds with the younger sample rather than with the older sample for three reasons: (1) it produces approximately equal age spans in each group (2 years each), and (2) 6-year-olds in the pilot data patterned more closely with 5-year-olds than with 7-year-olds. We will additionally fit a continuous-age model as a sensitivity check.

6.3 Cell-level chance tests

For each (`questionType` × `epistemic`) cell, we will conduct a Bayesian one-sample test of per-participant proportions against chance (0.5):

```
BayesFactor::ttestBF(x = pp$p_hypothesis, mu = 0.5, nullInterval = c(0.5, 1))
```

The directional `nullInterval = c(0.5, 1)` matches our directional predictions for H1 and H2. We will report the BF_{+0} (above-chance vs. null) and the 95% posterior credible interval on the cell proportion.

6.4 Age-stratified analyses

We will refit the §6.1 model separately within each age group, dropping the `ageGroup` term:

```
brm(hypothesisConsistent ~ questionType * epistemic
                         + (1 + questionType + epistemic | participantId),
    family = bernoulli(), ..., data = subset(dat, ageGroup == "Older"))
```

We will also report cell means and BFs (§6.3) stratified by age group.

6.5 Sensitivity analyses

1. **Continuous age.** Refit §6.1 with age centered and entered as a continuous predictor (replacing `ageGroup`).
2. **Prior sensitivity.** Refit §6.1 with (a) tighter priors (`Normal(0, 0.5)` on coefficients) and (b) wider priors (`Normal(0, 2.5)`).
3. **Trial-count threshold.** Refit §6.1 restricting to participants who completed all 12 trials.
4. **6-year-olds with the older group.** Refit §6.1 grouping 6-year-olds with 7–8 instead of 5–6, to check whether the age-grouping decision drives any inferential conclusions.
5.

6.6 Exploratory analyses (not confirmatory)

- Response-time analyses (log-RT as a continuous outcome) to check whether age and condition affect deliberation.
- Item-level random effects (random intercepts by `target` character).
- Effects of `rolesSwapped`, `blockOrder`, and `characterSet` as fixed effects (not predicted to matter; reported for transparency).
- 4 year olds

8. Materials and data availability

- **Stimuli and game code:** https://github.com/ashleyjthomas/RELKIND
- **Pre-registered analysis code:** `scripts/``analyze_twizzle_bayes.R`
- **Data:** raw trial-level data will be deposited on OSF upon project completion, with identifying fields (`firstName`, `chsId`) stripped.

9. References

Bürkner, P.-C. (2017). brms: An R package for Bayesian multilevel models using Stan. *Journal of Statistical Software, 80*(1), 1–28.